

LUT core-module ablation: three generations of lookup math

Comparison table

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Protocol

Architecture: LUT transformer, $E = 384$, 6 layers, 6 attention heads. FFN = **CompressionMultiHeadLUT** with $H = 4$ heads $\times$ $tph = 128$ tables ($P = H \cdot tph = 512$ tables per token), $d_{in} = d_{out} = 48$, $n = 8$ anchor pairs per table ($K = 2^n = 256$ cells); **compress** $384 \rightarrow 192$, **decompress** $192 \rightarrow 384$. The vanilla baseline (Table B row V) replaces this FFN with the standard dense one, $384 \rightarrow 1536 \rightarrow 384$, GELU, no bias. **Training:** 16,000 steps, batch size 48 (24576 tokens), untied embeddings, LUT tables without weight decay. **Eval:** **evaluate_bpb_fixed**, 48 rows $\times$ 100 steps, first 12 rows skipped.

Counting convention

- Core LUT only, per token, one layer. The **compress/decompress** projections are identical in every row and excluded (147,456 MACs forward).
- Row V has no LUT and no projections, so its figures are the whole dense FFN ($E \rightarrow 4E \rightarrow E$, $E = 384$); compare them with a LUT row plus the projections. Its 1,536 GELUs are not decomposed into FLOPs.
- **FLOP** = one scalar float operation ($+$, $-$, $\times$, $\div$, a comparison, $|\cdot|$, $\exp$, $\log \sigma$, σ). **MAC** = a multiply accumulated into a sum; 1 MAC = 2 FLOPs, included in the FLOP column. Memory reads and integer work are not counted.
- **Forward** = inference. **Backward** = all additional training arithmetic, including per-table gradients of per-layer scalars.

Table A — forward function and gradient function

Per variant, (a) is the function computed on the forward pass and (b) the function whose gradient training uses: the weights always receive the gradient of (a), while the input, the temperatures and the learned per-layer scalars receive the gradient of (b) with the table weights held constant.

Variant	(a) Forward: the function computed	(b) Backward: the function whose gradient is used
GEN 1 — Spiking Manifesto basic model		
1.1 Hard forward, two-alternatives backward	$y = \sum_t W_t[c_t]$, summed over tables t : each of table t 's n anchor pairs gives a signed margin $u_j = z[a_j] - z[b_j]$; the n signs $[u_j > 0]$ are the bits of the binary index $c_t \in \{0, \dots, 2^n - 1\}$ of the row read from $W_t \in \mathbb{R}^{K \times d}$, $K = 2^n$.	The 2-cell blend $\sum_t [(1 - U_t)W_t[c_t] + U_tW_t[c'_t]]$, $U_t = U(u_{j^*})$, $U(u) = 0.5/(1 + u)$, with the least-confident pair $j^* = \arg \min_j u_j $ and c' (c with bit j^* flipped) held fixed. The pairing with (a) is exact as a stop-gradient composite; the blend is C^1 across a bit flip and jumps only where j^* switches.
1.2 Soft forward, two-alternatives backward	(a) = (b): the blend of 1.1(b), applied at eval too.	
GEN 2 — FastMHL math		
2.1 Hard forward, all-alternatives backward (all K cells)	$y = \sum_t W_t[c_t]$, as 1.1(a).	With $b_j(\ell) = +1$ if the bit of ℓ for pair j is 1, else -1 , and the squashed margin $\hat{m}_j = u_j/(T_s + u_j) \in (-1, 1)$: $s_\ell = \sum_j b_j(\ell) \hat{m}_j$ scores how well row ℓ 's bit pattern agrees with the actual signs of the margins (small margins count little either way). All K rows of each table are read, softmax-weighted within that table: $\sum_t \sum_{\ell < K} \text{softmax}_\ell(s_\ell/T_{sel}) W_t[\ell]$. T_s, T_{sel} : per-layer learned scalars. Globally C^1 in the input; never uses the address, whose row is the one with the largest weight.
2.2 Hybrid_smooth two-alternatives forward, all-alternatives backward	Reads two cells, the addressed c_t and the alternative c'_t (c_t with the bit of the least-confident pair $j^* = \arg \min_j u_j $ flipped), blended by a weight set by that pair's margin: $y = \sum_t [(1 - w_t)W_t[c_t] + w_tW_t[c'_t]]$, $w_t = \sigma(-2\rho_{j^*}/T_{sel}) \in (0, \frac{1}{2}]$, $\rho_j = u_j /(T_s + u_j)$, σ the logistic sigmoid; applied at eval too.	The same softmax-weighted read of all K rows as 2.1(b), with the same T_s, T_{sel} parameters as (a).
2.3 Hard forward, 2-alternatives backward	The hard read of 1.1(a).	The softmax of 2.1(b) restricted to the two cells c_t and c'_t (c_t with the bit of the least-confident pair $j^* = \arg \min_j u_j $ flipped) and renormalised over that pair. The weight on c'_t is $\sigma(-2\rho_{j^*}/T_{sel})$, $\rho_j = u_j /(T_s + u_j)$, σ the logistic sigmoid: exactly 2.2(a)'s w_t , so this is 2.2(a)'s 2-cell blend.
2.4 Hybrid_smooth two-alternatives forward, 2-alternatives backward	(a) = (b): the 2-cell blend of 2.2(a). The backward keeps only 2 of the K rows, the addressed c_t and its least-confident neighbour c'_t , which are exactly the two cells the forward reads, so the restricted softmax equals the forward weight w_t and forward and backward coincide; unlike 2.2, the input gradient reaches only pair j^* .	
GEN 3 — LightMHL math (LookupFFN-inspired): learned margin score		
3.1 The Gen-3 form: learned margin score, one cell, per-layer (β, γ)	(a) = (b): $y = \sum_t s_t W_t[c_t]$ with the address c_t held fixed; $s_t = S_t P_t^\gamma$, $S_t = \sum_j m_j$, $P_t = \prod_j \sigma(\beta m_j)$, $m_j = u_j $, computed as $S_t \exp(\gamma \sum_j \log \sigma(\beta m_j))$; β, γ learned per layer from (2, 1), where s_t is LookupFFN's margin score. Discontinuous at cell boundaries, since $\sigma(0) > 0$.	
3.2 The Gen-3 form, two alternatives, per-layer (β, γ, τ)	(a) = (b): each table reads two cells, $y = \sum_t s_t [(1 - v_t)W_t[c_t] + v_tW_t[c'_t]]$, with s_t as in 3.1 over all n margins and c'_t the cell c_t with the bit of the smallest margin $j^* = \arg \min_j m_j$ flipped (c_t, c'_t, j^* held fixed). v_t is a softmax over the two cells with logits 0 and $-2m_{j^*}/\tau$ (flipping bit j lowers $\langle u, b \rangle$ by $2m_j$), i.e. $v_t = \sigma(-2m_{j^*}/\tau) \in (0, \frac{1}{2}]$; τ learned per layer from 0.5. The input gradient reaches all n margins through s_t and m_{j^*} alone through v_t . Continuous across a bit flip ($v_t = \frac{1}{2}$ there); jumps where j^* switches.	

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Variant	(a) Forward: the function computed	(b) Backward: the function whose gradient is used
3.3 Hard forward, 3.1 backward (one cell)	The hard read of 1.1(a): $y = \sum_t W_t[c_t]$, not scaled by a score; nothing else is computed at eval.	3.1's $\sum_t s_t W_t[c_t]$, $s_t = S_t P_t^\gamma$, $S_t = \sum_j m_j$, $P_t = \prod_j \sigma(\beta m_j)$, $m_j = u_j $, σ the logistic sigmoid, β, γ learned per layer. The input, β and γ get its gradient; the weights get (a)'s unscaled gradient at $W_t[c_t]$.
3.4 Hard forward, 3.2 backward (two alternatives)	The hard read of 1.1(a), as 3.3(a).	3.2's $\sum_t s_t [(1 - v_t)W_t[c_t] + v_t W_t[c'_t]]$, s_t as in 3.3(b), c'_t the cell c_t with the bit of the smallest margin $j^* = \arg \min_j m_j$ flipped, $v_t = \sigma(-2m_{j^*}/\tau)$, τ learned per layer. The input, β , γ and τ get its gradient; the weights get (a)'s unscaled gradient at $W_t[c_t]$ only.

Table B — cost and results

Per token, one layer, core LUT only (row V: the whole dense FFN); $P = 512$, $n = 8$, $K = 256$, $d = 48$, $E = 384$. `abl_NN` names a cell's prepared run folder in `experiments/ffn_replacement/lut_ablation`.

Variant	Forward FLOPs	Backward FLOPs	Forward MACs	Backward MACs	Validation bpb	Validation bpb, +TV ($\lambda = 10$)	Implementation (class used)
V	$2 \cdot 2E \cdot 4E = 2,359,296 + 1,536$ GELUs	$2 \cdot 4E \cdot 4E + 4E = 4,720,128 + 1,536$ GELU'	$2E \cdot 4E = 1,179,648$	$4E \cdot 4E = 2,359,296$	1.165147 (exp_n_0135); seed 2: 1.161798 (exp_n_0176); rerun abl_10	n/a: no LUT tables	Dense FFN (ffn_type='dense'): Linear(384, 1536, bias=False), GELU, Linear(1536, 384, bias=False).
1.1	$P(2n + d) = 32,768$	$P(5d + 2n + 9) = 135,680$	0	$P \cdot 2d = 49,152$	TBD — not yet run (abl_11)	TBD — not yet run (abl_12)	CompressionMultiHeadLUT(lut_impl='gen1'): MultiHeadLut(smooth_mode=False, n_alternatives=1, uncertainty_mode=INVERSE_L1, weights_init='uniform'), each head's anchors in its own slice.
1.2	$P(4d + 4n + 4) = 116,736$	$P(8d + 9) = 201,216$	$P \cdot 2d = 49,152$	$P \cdot 4d = 98,304$	TBD — not yet run (abl_13)	TBD — not yet run (abl_14)	As 1.1 with smooth_mode=True.
2.1	$P(2n + d) = 32,768$	$P(2Kd + 4nK + 9K + 11n + d) = 18,026,496^a$	0	$P(Kd + 2nK) = 8,388,608$	1.180622 (exp_n_0121); rerun abl_07	TBD — not yet run (abl_15)	FastMultiHeadLut(forward_mode='hard', backward_topk=0).
2.2	$P(2kd + 4n + 5) = 117,248$, $k = 2$ cells read	$P(2Kd + 4nK + 9K + 11n + 2kd) = 18,100,224^a$	$Pkd = 49,152$	$P(Kd + 2nK + kd) = 8,437,760$	TBD — not yet run (abl_01)	TBD — not yet run (abl_16)	FastMultiHeadLut(forward_mode='hybrid_smooth', backward_topk=0).
2.3	$P(2n + d) = 32,768$	$P(2kd + 4nk + 9k + 12n + d - 1) = 213,504$, $k = 2$ cells	0	$P(kd + 2nk) = 65,536$	TBD — not yet run (abl_02)	TBD — not yet run (abl_17)	FastMultiHeadLut(forward_mode='hard', backward_topk=1); backward_topk counts the cells kept besides c_t .
2.4	$P(2kd + 4n + 5) = 117,248$	$P(4kd + 9n + 15) = 241,152$	$Pkd = 49,152$	$P2kd = 98,304$	TBD — not yet run (abl_03)	TBD — not yet run (abl_18)	FastMultiHeadLut(forward_mode='hybrid_smooth', backward_topk=1).

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Variant	Forward FLOPs	Backward FLOPs	Forward MACs	Backward MACs	Validation bpb	Validation bpb, +TV ($\lambda = 10$)	Implementation (class used)
3.1	$P(2kd + 7n + 1) = 78,336$, $k = 1^b$	$P(4kd + 10n + 8) = 143,360^b$	$Pkd = 24,576$	$P2kd = 49,152$	1.169328 (exp_g_0248); rerun abl_08	1.163698 (exp_g_0249); rerun abl_09	LightMultiHeadLUT(confidence_form='learned_margin', learned_margin_init=(0, 2, 1), learned_margin_freeze_g=True), with g the log-gain, held at 0.
3.2	$P(2kd + 8n + 8) = 135,168$, $k = 2^b$	$P(4kd + 10n + 23) = 249,344^b$	$Pkd = 49,152$	$P2kd = 98,304$	TBD — not yet run (abl_04)	TBD — not yet run (abl_05)	As 3.1 with read_top_n=2, read_tau=0.5, read_tau_learnable=True.
3.3	$P(2n + d) = 32,768$	$P(3d + 15n + 9) = 139,776^b$	0	$Pd = 24,576$	TBD — not yet run (abl_19)	TBD — not yet run (abl_20)	As 3.1 with forward_mode='hard'.
3.4	$P(2n + d) = 32,768$	$P(5d + 16n + 29) = 203,264^b$	0	$P2d = 49,152$	TBD — not yet run (abl_21)	TBD — not yet run (abl_22)	As 3.2 with forward_mode='hard'.

Notes to Table B.

^a Dominated by $2Kd$: every table's K rows are read; several [tokens, 512, 256] fp32 buffers of about 12.9 GiB at 24,576 tokens.

^b Per table, forward: n comparisons, n differences, n absolute values, n products βm_j , $n \log \sigma$, the two sums, $\times \gamma$, exp and the final product $(7n + 1)$, plus the scaled rows $(2kd)$; 3.2 adds the argmin $(n - 1)$, the logit $-2m_{j^*}/\tau$ (2), the two-way softmax (4) and s_t times the two weights (2). Backward: the row terms $(4kd)$, the input term $(8n + 3)$ and the per-table accumulation of the parameter gradients ($\partial \log \gamma$ 3, $\partial \log \beta$ $2n + 2$); 3.2 adds ∂s_t from two rows (3), ∂v_t (2), the softmax backward (5), ∂m_{j^*} (3) and $\partial \log \tau$ (2). The parameter gradients are scalar multiplies, so they add no MACs. 3.3, 3.4: the forward is the hard read, so the score $(5n + 1)$, all of 3.1's forward but the comparisons and differences) and, for 3.4, the argmin, logit and softmax $(n + 5)$ are training-only and join 3.1's / 3.2's backward, which swaps each row's score-scaled weight update $(2d, d$ MACs) for one plain update at c_t (d): $3d + 15n + 9$ and $5d + 16n + 29$.

Cell TV regularisation (Table B's +TV column)

- **Penalty:** $\lambda \cdot \frac{1}{6} \sum_{\text{layers}} \frac{1}{Pn2^{n-1}} \sum_t \sum_{(c,c')} \|W_t[c] - W_t[c']\|^2$, squared L2 over the cell pairs (c, c') that differ in one bit, $\lambda = 10$; its gradient is added once per step, before gradient clipping. Training only: +6,144 FLOPs and +2,048 MACs per token per layer (amortised over 24,576 tokens); inference unchanged.
- **All three generations:** the trainer's penalty (lut_tv_penalty) covers MultiHeadLut, FastMultiHeadLut and LightMultiHeadLUT tables, whose Hamming-1 cell pairs were checked through each module's read path, and build_model refuses $\lambda > 0$ on a model whose tables it cannot reach.

Remaining work

- **Run the 21:** every cell above (20 LUT cells and V) is prepared, one folder each, in experiments/ffn_replacement/lut_ablation, whose README gives host and trainer; none is run yet. 2.1, 3.1, 3.1 +TV and V are reruns of their published runs, to put the whole table on one set of trainers.
- **Seeds:** zero replicates at this geometry; one extra seed of exp_g_0248 (≈ 0.95 h), prepared as abl_06, would give a real LUT seed spread. Reference: vanilla dense FFN 1.1651 / 1.1618 at 16K (two seeds), 1.1154 at 48K.